

Preference-Aware Coefficient Correction for Rewarded-Soups-Style Model Merging



Motivation and Goal

$$\theta_i \rightarrow \delta_i = \theta_i - \theta_0 \rightarrow R \rightarrow \lambda^* = f(p, R)$$

RS: train m specialists, serve any preference p by interpolating weights

RS sets $\lambda = p$, ignoring how the specialists overlap

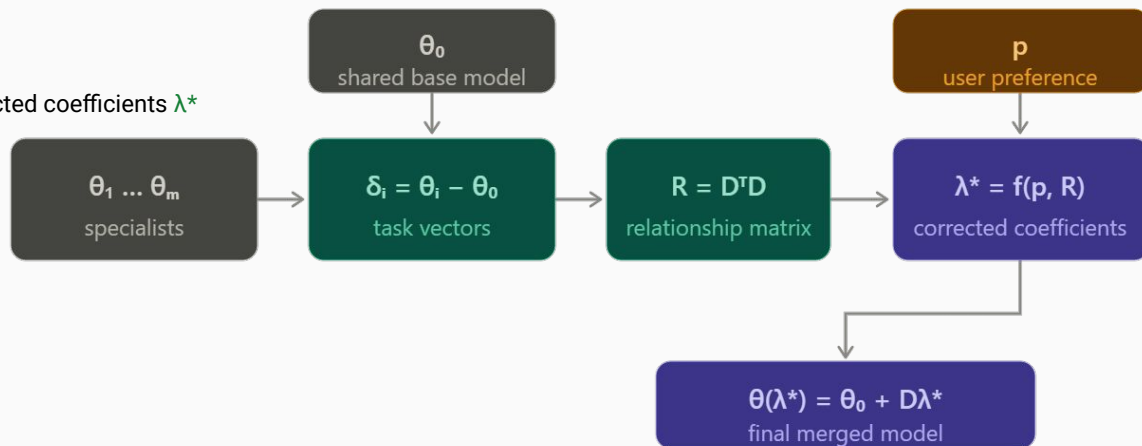
R is computed from the adapter weights alone: zero reward queries at merge time, no retraining

But R measures overlap in parameter space. Whether that predicts reward behaviour is what this thesis tests

Both answers are results: a free improvement, or an applicability boundary

1. Train attribute adapters
2. Extract task vectors
3. Compute relationship matrix R
4. Given preference vector p , solve for corrected coefficients λ^*
5. Merge adapters
6. Evaluate merged model
 - a. Linear utility:

$$U_p(\theta) := \sum_{i=1}^m p_i r_i(\theta)$$



Architecture Overview

- Adapter $\delta \in \mathbb{R}^P$

$$D = [\delta_1, \dots, \delta_m]$$

- $D \in \mathbb{R}^{(P \times m)}$
- Setup: TinyLlama-1.1B, five HelpSteer2 axes, RS-PPO specialists, ~50 GPU h
- Compute relationship matrix:
 - $R \in \mathbb{R}^{(m \times m)}$
- Compute corrected coefficients:
 - $\lambda \in \mathbb{R}^m$
- Merge adapters and evaluate with ArmoRM

$$R = D^\top D$$

$$\lambda^* = f(p, R)$$

$$U_p(\theta) := \sum_{i=1}^m p_i r_i(\theta)$$

Methods

$$\mathbf{Avg} : \quad \lambda_{\text{Avg}} = \arg \max_{\lambda \in \Delta_m} p^\top R \lambda - \rho (\lambda - p)^\top R (\lambda - p),$$

$$\mathbf{Cert} : \quad \lambda_{\text{Cert}} = \arg \max_{\lambda \in \Delta_m} \min_i \Delta_i(\lambda) \quad \text{s.t.} \quad (\lambda - p)^\top R (\lambda - p) \leq c^2 \max(p^\top R p, \varepsilon),$$

$$\mathbf{MaxMin}(c) : \quad d^\star = \arg \max_{\mathbf{1}^\top d = 0} \min_i (Rd)_i \quad \text{s.t.} \quad \|d - d_0\|_R \leq c \|d_0\|_R, \quad d_0 = \Pi_0(Rp),$$

$$\lambda_{\text{MaxMin}(c)} = p + d^\star,$$

$$\mathbf{Fair}(\alpha, \varepsilon) : \quad \lambda_{\text{Fair}} = \arg \max_{\lambda \in \mathcal{F}_p^\varepsilon} U_\alpha(\Delta(\lambda) + \varepsilon \mathbf{1}) \quad \text{s.t.} \quad (\lambda - p)^\top R (\lambda - p) \leq c^2,$$

Training

- SFT — Supervised Fine-Tuning
 - highly shared quality direction
- PPO — Proximal Policy Optimization
 - most closely aligned with the original Rewarded Soups paper
- DPO — Direct Preference Optimization
 - produced almost orthogonal parameter updates

RS-PPO

- RS-PPO evaluated 137 unique merge points across 126 preregistered comparisons
- Overall, parameter geometry did not reliably predict external reward improvements

DPO

$$R_{\text{DPO}} \approx \begin{pmatrix} 1 & .0644 & .0511 & .0153 & .0073 \\ .0644 & 1 & .0437 & .0172 & .0039 \\ .0511 & .0437 & 1 & .0167 & .0005 \\ .0153 & .0172 & .0167 & 1 & .0399 \\ .0073 & .0039 & .0005 & .0399 & 1 \end{pmatrix}$$

EV: 0.934, 0.956, 0.962, 1.032, 1.116

Results

- DPO produced almost orthogonal adapter updates, with cosine similarities between 0.001 and 0.064
- None of the 226 method comparisons remained significant after Holm correction
- Overall, distinct parameter directions did not translate into reliable functional improvements

Next steps

- Work on main.tex
- Repeat the experiments across multiple random seeds
- Try to improve expert specialization